Student Enrollment & Performance Dashboard

Analysis Report

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Executive Summary

This report analyses student enrollment, scores and attendance across four academic programs and ten courses. The results are presented in a two-page Power BI dashboard: an Overview page and a Performance Analysis page. After cleaning, the data holds 30 registered students and 50 course enrollments.

The main findings are:

- **Enrollment:** 50 enrollments came from 25 of the 30 registered students. Five students have not enrolled in any course.
- **Performance:** the average score is 76.98 and the pass rate is 94% (47 of 50 enrollments).
- **Programs:** Data Analytics students average 88.4, about 18 points above Information Technology students at 70.0.
- **Failures:** all three failed enrollments belong to male students, in Python Programming, Web Technologies and Information Systems.
- **Course uptake:** Introduction to Data Analytics is the most popular course (7 enrollments). SQL Fundamentals has only 1, and Data Visualization has none.

1. Introduction and Objectives

The task was to act as a data analyst for an education provider, using three datasets (Enrollment, Student and Course) to build a Power BI dashboard. The dashboard had to show:

1. Number of enrollments by course
2. Average score by course
3. Number of students by program
4. Total enrollments
5. Total students
6. Average score

I added supporting visuals (gender, enrollment status, enrollments over time, scores by instructor, students by city and a student-level detail table) to give the required numbers more context.

2. Data Description

- **Enrollment:** one row per student-course enrollment, with Enrollment_ID, Student_ID, Course_ID, Enrollment_Date, Score, Attendance_Rate and Status. It had 52 rows before cleaning.
- **Student:** 30 students with Student_ID, Student_Name, Gender, Age, City and Program.

- **Course:** 10 courses with Course_ID, Course_Name, Department, Instructor and Course_Level.

Each enrolled student takes two courses within their own program's department, which is why the same Student_ID appears twice in the Enrollment table. This is not duplication. The structure also gave a reliable rule for recovering missing values.

3. Data Cleaning and Preparation

All cleaning was done in Power Query so that every step is recorded, repeatable and non-destructive. The raw data was left untouched. The table below summarises the issues found and how each was handled.

Issue	Records	Treatment
Fully duplicated rows	EN047, EN048	Removed, leaving 50 enrollments
Missing Enrollment_ID	6 rows	Regenerated from the row sequence after de-duplication, validated against the existing IDs, and flagged as Inferred
Missing Student_ID (Enrollment)	EN004, EN014	Set to ST002 and ST007 from each student's course pairing and the consecutive IDs
Missing Course_ID	EN035	Set to C004, because the student is in Computer Science and the paired enrollment is C005
Missing Student_ID (Student table)	5 students	Recovered as ST005, ST008, ST011, ST014 and ST016, cross-checked against each student's courses and program
Missing Program	Grace Eze	Set to Data Analytics from her course enrollments
Missing Status	EN008, EN014	Derived from score: Failed (score 40) and Passed (score 69)
Missing Score	EN029	Left blank, so it is excluded from averages
Missing Enrollment_Date	EN001, EN006, EN032	EN032 inferred as 25 January 2026 from the consistent 6-day gap between a student's two courses, and flagged. The other two were left blank
Missing Gender or City	1 gender, 3 cities	Set to Unknown

Inferred values were flagged in separate quality columns (ID_Source, Date_Quality and Score_Quality) so that they can be identified and excluded if required. A pass mark of 60 was assumed, because the data shows Failed at 58 and Passed at 64.

4. Data Model and Measures

The model is a star schema. Enrollment is the fact table, linked one-to-many from the Students table (Student_ID), the Courses table (Course_ID) and a Dim_Date calendar table (Date to Enrollment_Date_Clean).

The main DAX measures are:

- **Total Enrollments** = COUNTROWS(Enrollment)
- **Total Students** = DISTINCTCOUNT(Students[Student_ID])

- **Enrolled Students** = DISTINCTCOUNT(Enrollment[Student_ID])
- **Avg Score** = AVERAGE(Enrollment[Score]), which ignores the blank score
- **Pass Rate** = passed enrollments divided by total enrollments

5. Dashboard Design

The dashboard uses a single-hue dark blue palette, so the colour is consistent and the data stands out. The page background is #285D8E, the main panel is #082F51, and cards and chart boxes are #0B3E65. Charts use light blue (#9DB2C4) with pale blue text for contrast. A left sidebar provides page navigation.

Page 1: Overview. The page has five KPI cards (Total Enrollments, Total Students, Average Score, Enrolled Students and Pass Rate), each with an icon and a small trend line. Total Students has an enrolled versus not-enrolled bar instead, because students have no date. Below the cards are Enrollments by Course, Gender Distribution, Enrollment by Status, Enrollments over Time, Average Score by Course and Students by Program. A program slicer sits in the header.

Page 2: Performance Analysis. The page has Average Score by Program, Students by City, Average Score by Instructor, and an Enrollment Details table. A Selected Student Record panel shows the full record for whichever row is clicked, and dropdown slicers filter by Instructor, Program and City.

6. Key Findings

6.1 Overall picture

There were 50 enrollments from 25 of the 30 registered students, so 83% of students are enrolled. The five students who have not enrolled are Victor Udo, Joy Okeke, Abdul Rahman, Linda George and Oluwaseun Adebayo. The average score is 76.98, calculated from 49 scores, and the pass rate is 94%, with 47 enrollments passed and 3 failed.

6.2 Enrollments and scores by course

Introduction to Data Analytics has the most enrollments (7). Seven courses have 6 each, and SQL Fundamentals has just 1. Data Visualization has none, so it does not appear on the chart.

Average scores range from 90 for Introduction to Data Analytics and 88 for Power BI for Reporting down to 71 for Database Management and 67 for Information Systems. The two Data Analytics courses at the top stand about 14 to 23 points above the weakest courses.

6.3 Performance by program

Data Analytics averages 88.4, Business Administration 74.5, Computer Science 72.9 and Information Technology 70.0. Students are spread evenly across the programs: Data Analytics and Computer Science have 8 each, and Business Administration and Information Technology have 7 each.

6.4 Enrollment status and gender

Of the 50 enrollments, 47 (94%) passed and 3 (6%) failed. The three failures are Michael Ade in Information Systems (score 40), Yusuf Abdullahi in Python Programming (58) and John Mark in Web Technologies (55). All three are male.

By student count the split is 15 female, 14 male and 1 unknown. Female students average about 82.3 and male students about 71.3. This gap needs care when interpreting. Gender and program are almost completely aligned in this dataset: all 8 Data Analytics and all 7 Business Administration

students are female, and all 7 Information Technology students are male, as are 7 of the 8 Computer Science students. The gender difference therefore largely reflects the program difference described above, and the data cannot separate the two.

6.5 Instructors

Average scores by instructor run from 90.3 (Dr. Ade) and 88.3 (Mr. Musa) down to 71.3 (Mrs. James) and 66.5 (Mrs. Adeyemi). In this dataset each instructor teaches only one course, so the instructor chart repeats the course chart and cannot show whether instructor quality, course difficulty or the student cohort explains the differences. It should not be used to judge individual instructors.

6.6 Enrollments over time

Enrollment dates run from 8 January to 3 February 2026. Weekly enrollments were 9 (week of 5 Jan), 13 (12 Jan), 14 (19 Jan), 10 (26 Jan) and 2 (2 Feb). The final week is partial because the data ends on 3 February. Two enrollments with no recorded date are excluded from this chart.

6.7 Student locations

Students come from ten cities. Lagos, Abuja, Ibadan and Enugu have 4 students each. Kano, Kaduna and Unknown have 3 each, Owerri and Port Harcourt have 2 each, and Uyo has 1.

7. Recommendations

1. **Review the low-uptake courses.** SQL Fundamentals has one enrollment and Data Visualization has none. Check whether they are scheduled, promoted or required, and consider whether they should be offered or merged.
2. **Follow up the five students who have not enrolled.** They make up 17% of registered students and may need guidance or enrollment support.
3. **Provide extra support in the three courses with failures.** Python Programming, Web Technologies and Information Systems each had one failed student, and Information Systems has the lowest average score (67).
4. **Investigate the program gap.** Information Technology averages about 18 points below Data Analytics. Collect more data (for example, course difficulty and prior preparation) before drawing conclusions.
5. **Improve data capture.** Missing IDs, dates, genders, cities and programs had to be inferred or left blank. Making these fields mandatory at entry would remove that risk.

8. Assumptions and Limitations

- **Pass mark:** a score of 60 was assumed. The original status labels were kept wherever they were present.
- **Inferred values:** the missing IDs, one enrollment date and one program were inferred from patterns in the data. They are flagged, but they are inferences rather than recorded facts.
- **Excluded data:** one enrollment has no score and is excluded from averages. Two enrollments have no date and are excluded from the time trend. All are retained in other counts.
- **Small sample:** with 30 students and 50 enrollments, the figures describe patterns and do not support statistical conclusions. A single student can move a program average noticeably.

- **Confounded groups:** program, gender, course and instructor overlap heavily in this dataset, so differences between them cannot be attributed to any single factor.
- **Short period:** the data covers about four weeks, and the last week is partial, so the trend reflects an early part of the term only.

9. Conclusion

The dashboard answers all six required questions and adds context through supporting visuals. Enrollment is broad, with 25 of 30 students enrolled, and the pass rate of 94% is high. The main areas for attention are the five students who have not enrolled, the weakest-performing courses and the gap between programs. The data quality fixes are documented, so every figure in the dashboard can be traced to a cleaning step.